ABCPROOF · MOTIZUKIPDF · STRATEGY-GUIDE REVIEW

The Gokun Strategy Guide

We read Go Yamashita's 409-page survey *A proof of the abc conjecture after Mochizuki* and marked our own homework against it: what it taught us, what it confirmed, and what in this repository it shows to be wrong or sloppy. With figures, real numbers, and red-pen stamps. （山下サーベイで答え合わせ：攻略本を読んでみた）

Paper: version of June 25, 2024, now in `pdf/` · examples: the 4 sets of initial Θ-data the toolkit builds exactly · every example number comes from `iut pipeline` · November 2026

ABSTRACT Until now this project read Mochizuki in the original: four IUT papers, [Alien], [EssLgc]. Yamashita's survey [Y] is the 攻略本, the thick strategy guide you buy after getting stuck on the same boss for a week. It re-proves the whole route from anabelian geometry to the abc inequality in one self-contained document, with its own notation and with the elementary computations written out in full. Reading it produced three kinds of result. **CORRECTIONS:** four places where our toolkit or earlier articles say something [Y] shows to be wrong or empty (§2). **CONFIRMATIONS:** eight conventions we had chosen that match (§2). **NEW MATERIAL:** seven ideas that none of our earlier write-ups contain, including the "miracle identity", the reason the log-theta-lattice needs infinite columns, and a short proof that the obvious cheat code Θ^N cannot work (§3–§9). The four corrections have since been made in the toolkit and in `funfun_iut.pdf`; §10 says what changed.

CONTENT NOTICE · ご注意

All pop-culture references are text only: no character art, logos or screenshots. The stamps and diagrams are drawn for this article. Where we say what the survey proves, we cite its numbering. The survey is an exposition by a RIMS mathematician who thanks Mochizuki for many discussions; it presents the proof as complete and does not discuss the 2018 objection of Scholze and Stix (a text search of all 409 pages finds neither name, nor Joshi's). As in our earlier guides, we report and adjudicate nothing (§9).

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4 The level cap: our examples are below it
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1 The book on the table 机の上の一冊

The file that arrived was the text layer of `abc2024Jun25.pdf`. The same document is public on the author's RIMS page, so the PDF itself, 409 pages, is what went into `pdf/`. Its first line of content is the abstract, which is two sentences long. The remaining 409 pages are the fine print.

The author's e-mail handle is *gokun*, and we will make the obvious joke exactly once: this is a man who went to train under a master on a remote mountain (Kyoto has several), ran an intensive seminar on the technique from May to November 2013, and came back with a 409-page scroll. The acknowledgments thank the executives of Toyota Central R&D Labs for "a special position that enabled him to concentrate on his re-

search in pure mathematics”, and thank Sakichi Toyoda, who died in 1930, for the philanthropic culture behind it. This makes [Y] the only survey of the abc conjecture with an automotive sponsor, and explains its method, which is *kaizen*: no single step is hard, there are simply a great many of them and none is skipped.



Figure 1. Page budget of [Y], from the running heads of the PDF. The part that mentions heights, conductors and the abc conjecture at all is §1, 34 pages, 8% of the book. The index alone is 28 pages.

The introduction opens with a story attributed to Grothendieck: there are two ways to crack a nut. One is a nutcracker. The other is to soak it in water “for an extended period of time until its shell dissolves naturally”. Japanese cuisine has a third way, which is to bury it in rice bran for three years and call the result 糠漬け, and that is closer to what happens here. Yamashita then says something none of our earlier write-ups dared to:

> *“...these numerous trivial constructions, whose exposition occupies literally hundreds of pages, allow one to conclude a highly nontrivial consequence ... practically effortlessly!”*

and, a paragraph later, that if one sets aside the delicate way the pieces are combined, “the only nontrivial mathematical ingredient in inter-universal Teichmüller theory is the classical result [pGC], which was already known in the last century”. [pGC] is Mochizuki's 1999 theorem on the pro-p anabelian geometry of curves, the subject of our `anabelian_guide.pdf`. In anime terms the survey is describing a slice-of-life series: for eleven episodes the cast makes tea, labels prime-strips and walks to school, and then in episode twelve somebody says one sentence and you find you are crying. The sentence is Corollary 13.13. The tea is §2–§12.

OTAKU CORNER · 攻略本あるある

Every strategy guide has the same three features. A world map on the inside cover: here it is the *Leitfaden* on page 9, with the honest caption that the dependences are “rough (or conceptual)”. A bestiary at the back: here Appendix C.6, pages of tables listing every model object, bridge and theatre. And one page where the authors admit which side quests you can skip. [Y] §0.1 has two: the archimedean theory “does not play a very central role”, and the main reconstruction theorems of the theory of Frobenioids “do not play a very central role” either. Our toolkit implements both in loving detail. We regret nothing.

2 The report card 通知表

Japanese teachers mark homework with round red stamps. We made four: 訂正 **correction**, something of ours is wrong; 注意 **caution**, something of ours is true but misleading; 合格 **pass**, a convention we chose matches the survey; 新知 **new**, not in any earlier write-up of ours. Every row cites where in [Y] to look.

	what we had	what [Y] says	where
訂正	`funfun_iut.pdf` §1 said: $\log q$ is "the arithmetic degree of the q-parameters at the bad places v_2, v_3, v_5, v_7".	$\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}$ never contains a place over 2. For $1 + 2400 = 2401$ the number we printed, $24.2023 = 2\log 3 + 4\log 5 + 8\log 7$, is right and uses only v_3, v_5, v_7; the sentence is wrong. The place 2 counts only in $\log q^{\forall} = 25.5886$.	Def. 10.1 (3); §1.3 (3); proof of Prop. 1.15
訂正	The toolkit printed ramification index $e_K = 1$ at every place, including the bad ones, and built log-shells as $p^0 \cdot \mathcal{O}$.	Impossible at a bad place: $q_v^{1/2l} \in K_v$, because $K = F(E[l])$ and $E[2]$ is rational. So $e(K_v/\mathbb{Q}_p)$ is a multiple of $2l/\gcd(2l, \operatorname{ord} q_v)$. See §2.1.	Lemma 1.8; proof of Lemma 1.9 (1)
訂正	`ComputeThmOneTen` and `ComputeCorTwoTwo` ended with a bare "holds" for every example.	Prop. 1.15 (Cor. 2.2) is proved only *outside a finite exceptional set*, and all our examples are inside it. Prop. 1.12 (Thm. 1.10) has no exceptional set, but for these curves it holds through its constant alone: "holds" meant $4 \le 10^{22}$. See §4.	Prop. 1.12; Prop. 1.15, step (s1) and the last enlargement of Exc
訂正	`iut pipeline` stage 1 printed √ for item (a), "F/F_{mod} Galois of degree prime to l", even for $l = 5$, while `iut validate` on the same data reported the error.	The condition is part of the definition, and in the proof $l = 5$ never occurs: $5 \le \sqrt{h} < l$. Found while checking this against the code; stage 1 reads only the asserted flag.	Def. 10.1 (2); Prop. 1.15 (P1)
注意	`funfun_iut.pdf`: IUT "turns the triple into the Frey curve and proves a Szpiro / Vojta type inequality".	There are two more reductions in between, and the Frey curve of *your* triple is not what the theory is applied to in general. See §3.	Prop. 1.2, Prop. 1.15
注意	The GUI's **auto l** and the Θ-data generator choose "the smallest prime $l \ge 5$" that passes Def. 3.1.	Legal as initial Θ-data, but not how the proof chooses: l is taken in a window just above $\sqrt{h}$ so that the two sides of the miracle identity balance. See §5.	Prop. 1.15 (P1)–(P3)
合格	$\underline{\epsilon}$ is a *non-zero* cusp, label 1 (decided in the pipeline thread).	"$\underline{\epsilon}$ is a non-zero cusp of the hyperbolic orbicurve $\underline{C}_K$".	Def. 10.1 (7)
合格	Log-shell $\mathcal{I}_v = (p_v^*)^{-1}\log(\mathcal{O}^\times)$, $p^* = p$ or 4.	$\mathcal{I} = \frac{1}{2p}\log_p(\mathcal{O}^\times)$: the same lattice (2 is a unit for odd p), in one formula.	§1.3; §5.1
合格	Θ-pilot log-volume $-\frac{l(l+1)}{12}\,\lvert\log q\rvert$.	$-\frac{l+1}{24}\log(q)$, which is the same thing since $\lvert\log q\rvert = \frac{1}{2l}\log q$.	Lemma 1.10 (1)
合格	$F = \mathbb{Q}(\sqrt{-1}, E[2\cdot 3\cdot 5])$, $e_{\mathrm{mod}}^* = 2^{12}\cdot 3^3\cdot 5\cdot e_{\mathrm{mod}} = 552960\, e_{\mathrm{mod}}$.	Same field, same constant.	§1.3 (1)–(2); Prop. 1.15
合格	Final form $\frac{1}{6}\log q \le (1 + \frac{20 d_{\mathrm{mod}}}{l})(\log \mathfrak{d}^{F_{\mathrm{tpd}}} + \log \mathfrak{f}^{F_{\mathrm{tpd}}}) + 20(e_{\mathrm{mod}}^* l + \eta_{\mathrm{prm}})$.	Identical, with the intermediate constants $1 + \frac{12 d_{\mathrm{mod}}}{l}$ and $10(\dots)$ that our C_Θ uses.	Prop. 1.12
合格	The validator warns that $l = 5$ is excluded when $F \supseteq \mathbb{Q}(E[5])$.	Agrees: $l > \sqrt{h} \ge 5$.	Prop. 1.15 (P1)
合格	Archimedean log-shell: a disc of radius π (the *alien rings* infographic).	$\mathcal{I}_{v_1,\dots,v_n} := \pi^n B$, and the archimedean cost per packet is $\le (j+1)\log\pi$.	§1.3; Lemma 1.9 (3)

	what we had	what [Y] says	where
合格	`multiradial_guide.pdf` §7: (Ind1), (Ind2) leave log-volumes alone, (Ind3) gives "$\leq$".	Same three, under different names; see the phrasebook in §10.	Thm. 13.12; Rem. 13.13.1
合格	The log-theta-lattice does not commute.	With a one-line reason we did not have: §6.	start of §13

2.1 Correction two, with numbers: the ramification we forgot

In the proof of Lemma 1.9, Yamashita needs $\lambda = \frac{j^2}{2l}\,\mathrm{ord}(q_{v_j})$ to lie in $\frac{1}{e_{v_j}}\mathbb{Z}$, and notes in passing why it does: $q_v^{1/2l} \in K_v$. Read backwards, that is a lower bound on the ramification of K at each bad place, and our data violated it everywhere (the last column is what the toolkit prints now):

example	place	$\mathrm{ord}_v(q_v)$	l	$\mathrm{ord}(q_v^{1/2l})$	smallest possible $e(K_v/\mathbb{Q}_p)$	toolkit's e_K, before	now
1 + 8 = 9	v_3	4	7	2/7	7	1	≥ 14
1 + 2400 = 2401	v_3	2	11	1/11	11	1	≥ 22
1 + 2400 = 2401	v_5	4	11	2/11	11	1	≥ 44
1 + 2400 = 2401	v_7	8	11	4/11	11	1	165
$2 + 3^{10}\cdot 109 = 23^5$	v_3	20	7	10/7	7	1	≥ 14
$2 + 3^{10}\cdot 109 = 23^5$	v_{23}	10	7	5/7	7	1	21
$2 + 3^{10}\cdot 109 = 23^5$	v_{109}	2	7	1/7	7	1	105
curve 11a1	v_{11}	5	7	5/14	14	1	42

The pipeline used to print $\mathrm{ord}(q_v) = 2/7$ and $e_K = 1$ on adjacent lines of the same run, which is the mathematical equivalent of a character whose age is listed as 17 and whose backstory takes 30 years. The log-volume totals survive, because they are normalised by degree, but two downstream objects do not: the log-shell at a ramified place is not $\mathcal{O}$ (Lemma 1.6 gives the true size, with the term $4(j+1)\,\iota\,\log(e^*_{\mathrm{mod}}l)$ of Lemma 1.9 (1) paying for it), and $\log \mathfrak{d}^K$ cannot be read off an unramified model. For the latter [Y] gives a rigorous ceiling to replace our estimate. Lemma 1.8 says K/F_{tpd} is unramified outside $2\cdot 3\cdot 5\cdot l$ and the support of q, with F/F_{tpd} tame outside $2\cdot 3\cdot 5$ and K/F tame outside l, whence Lemma 1.11 (1):

example	l	toolkit's estimate of $\log \mathfrak{d}^K$	Lemma 1.11 (1): $\log \mathfrak{d}^{F_{\mathrm{tpd}}} + \log \mathfrak{f}^{F_{\mathrm{tpd}}} + 2\log l + 21$	inside the bound?
1 + 8 = 9	7	5.684	25.990	yes
1 + 2400 = 2401	11	10.143	30.450	yes
$2 + 3^{10}\cdot 109 = 23^5$	7	13.510	33.817	yes
curve 11a1	7	6.290	27.290	yes

Our estimates sit inside the bound, so nothing published is contradicted. The bound, not the estimate, is what the proof uses.

3 The route has three warp pipes we skipped ワープ土管

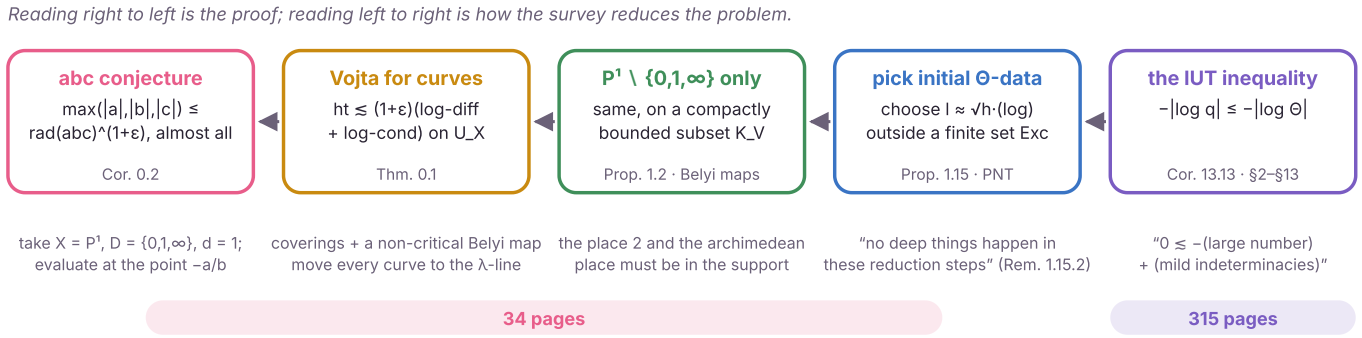


Figure 2. The whole proof as [Y] organises it. Our earlier articles drew only the first and last boxes.

Our earlier description was: take $a + b = c$, form the Frey curve, apply IUT. The survey's §1 shows that this compresses three separate reductions, each of which changes what object the theory is applied to.

First pipe (Cor. 0.2 from Thm. 0.1). abc is the case $X = \mathbb{P}^1$, $D = \{0, 1, \infty\}$, $d = 1$ of Vojta's conjecture for curves, evaluated at the rational point $-a/b$. The proof is a dozen lines and uses nothing but $|a + b| \leq 2\max\{|a|, |b|\}$. This is the tutorial level.

Second pipe (Prop. 1.2, from [GenEll]). To prove Vojta for *every* hyperbolic curve it is enough to prove it for $\mathbb{P}^1 \setminus \{0, 1, \infty\}$, and only on a *compactly bounded subset*: points that stay a fixed distance from the three cusps at finitely many chosen places. The tool is a non-critical Belyi map (Appendix C.2), a map to $\mathbb{P}^1$ branched only over $\{0, 1, \infty\}$ that sends one chosen finite set of points to the cusps while keeping another chosen finite set away from them. This is the step that matters for us. A high-quality abc triple is, by its nature, a point *close* to a cusp at some place; the theory is not asked to handle it head-on. The Belyi map carries the problem somewhere else first, like the green pipe that drops you in a different world with the same coin count.

Third pipe (Prop. 1.15). On a compactly bounded subset whose support contains the place 2 and the archimedean place, every point outside a finite set comes, after enlarging the field as in §2, from initial Θ-data. The archimedean place and 2 are awkward for the theory, and this is how they are neutralised: the height contribution there is bounded by a constant $C_{\mathcal{K}}$ depending only on the subset. That is why $\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}$ excludes 2 (correction one), and why nobody minds.

Only then does the fourth box, the one all the excitement is about, get used. Yamashita is blunt about the first three (Remark 1.15.2): the inequality $-|\log(q)| \leq -|\log(\Theta)|$ "is almost a tautological translation of the inequality which we want to show", and "no deep things happen in these reduction steps".

OTAKU CORNER · 逆から読む

The survey is organised like a mystery told in reverse. Chapter 1 is the culprit's confession ("it was the log-volume, in the log-shell, with the theta values"). Chapters 2 to 13 are the 315-page flashback that establishes whether the confession is admissible. The little detective with the bow tie would be done in twenty minutes plus commercials, but he has never had to reconstruct a number field from its Galois group.

4 The level cap: our examples are below it レベル不足

Proposition 1.15 does not say the inequality holds for every elliptic curve. It says there is a finite exceptional set $\mathrm{Exc}_{\mathcal{K},d,\epsilon}$, enlarged several times during the proof, outside which it holds. Two of the enlargements are explicit. Writing $h = \log(q^{\forall})$ and $d^* = 2^{12} \cdot 3^3 \cdot 5 \cdot d$:

- **(s1)** all curves with $h^{1/2} < \xi_{\mathrm{prm}} + \eta_{\mathrm{prm}}$ are thrown into Exc, where $\xi_{\mathrm{prm}} \geq 5$ and η_{prm} come from the prime number theorem;
- **the last step** throws in all curves with $\epsilon_E := (60d^*)^2\, h^{-1/2} \log(2d^*h) > \min\{1, \epsilon\}$.

[Y] keeps η_{prm} abstract. Our toolkit plugs in $5 \cdot 10^{20}$, citing [ExpEst]. With that value and $d = 1$:

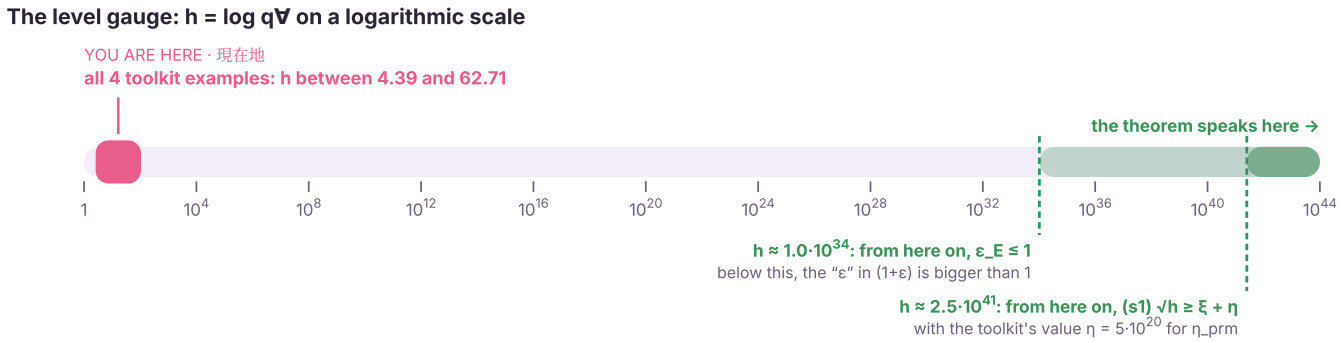


Figure 3. The condition $\epsilon_E \leq 1$ alone needs $h \gtrsim 1.0{\cdot}10^{34}$. Since $\frac{1}{6}h$ is the height up to a bounded error, that is an abc triple whose c has on the order of $7{\cdot}10^{32}$ decimal digits; condition (s1) with the toolkit's constant asks for about $2{\cdot}10^{40}$.

example	l	$h = \log q^{\forall}$	$\sqrt{h}$ (must be $< l$)	ε_E (must be $\leq \min\{1, \varepsilon\}$)	toolkit said	toolkit says now
1 + 8 = 9	7	4.39	2.10	$8.1{\cdot}10^{15}$	"holds"	in the exceptional set
1 + 2400 = 2401	11	25.59	5.06	$3.7{\cdot}10^{15}$	"holds"	in the exceptional set
$2 + 3^{10}{\cdot}109 = 23^5$	7	62.71	7.92	$2.5{\cdot}10^{15}$	"holds"	in the exceptional set
curve 11a1	7	11.99	3.46	$5.2{\cdot}10^{15}$	"holds"	in the exceptional set

So the toolkit's old "holds" was true and said nothing: the right-hand side it compares against contains $20(e^*_{\mathrm{mod}} l + \eta_{\mathrm{prm}}) \approx 10^{22}$, and the "$\epsilon$" in $(1 + \epsilon)$ is at best $2.5{\cdot}10^{15}$. This is correction three. It is not a defect of the theory, which only ever promised a statement about all but finitely many curves; it was a defect of our printout, which presented a comparison with a twenty-two-digit constant as a result. One precision, which the first version of this article blurred: the exceptional set belongs to Prop. 1.15 (Cor. 2.2). Prop. 1.12 (Thm. 1.10) is stated for every initial Θ-data, and for our curves it is true for the dull reason that its constant term alone exceeds the left side. The toolkit now says both things in so many words.

OTAKU CORNER · レベル上げ

Every JRPG has the door in the first village that says "you are not yet strong enough". Our four examples are standing in front of it at level 3, holding a wooden sword and the triple $1 + 8 = 9$. The door opens around level 10^{33}. There is no amount of grinding on slimes that gets you there, and the gacha does not help either: the drop rate of an abc triple with 10^{33} digits and quality above 1 is the thing the conjecture is about.

5 The miracle identity 奇跡の等式

Why is the constant in front of $(\text{log-diff} + \text{log-cond})$ exactly 1, and not 2, or 6, or π? None of our earlier write-ups asked. Remark 1.15.3 answers, and Yamashita names it himself: the *miracle identity*.

The bound on the Θ-pilot (Lemma 1.10) has two competing terms. The theta values q^{j^2} shrink the region: averaged over $j = 1, \dots, l^*$ they contribute $-\frac{l+1}{24}\log q$. The log-shells that hold them grow it: packet j has $j+1$ tensor factors, each costing one $\log \mathfrak{d}^K$, and the average is $\frac{l+5}{4}\log \mathfrak{d}^K$. Writing the height as $\frac{1}{6}\log q$,

$$-|\log(\Theta)| \;\le\; \underbrace{\frac{l+1}{4}}_{\text{quadratic, averaged}} \left(-\tfrac{1}{6}\log q\right) + \underbrace{\frac{l+5}{4}}_{\text{linear, averaged}} \log \mathfrak{d}^K + \cdots$$

and the two coefficients agree to leading order in l. Yamashita factors each into its causes:

$$\underbrace{6}_{\lambda\text{-line over } j\text{-line}} \times \underbrace{\tfrac{1}{2}}_{\text{theta lives on a double cover}} \times \underbrace{\tfrac{1}{2^2}}_{\text{exponent } j^2} \times \underbrace{\tfrac{1}{3}}_{\sum j^2 \approx n^3/3} = \underbrace{\tfrac{1}{2}}_{\text{differents are linear}} \times \underbrace{\tfrac{1}{2}}_{\sum j \approx n^2/2}.$$

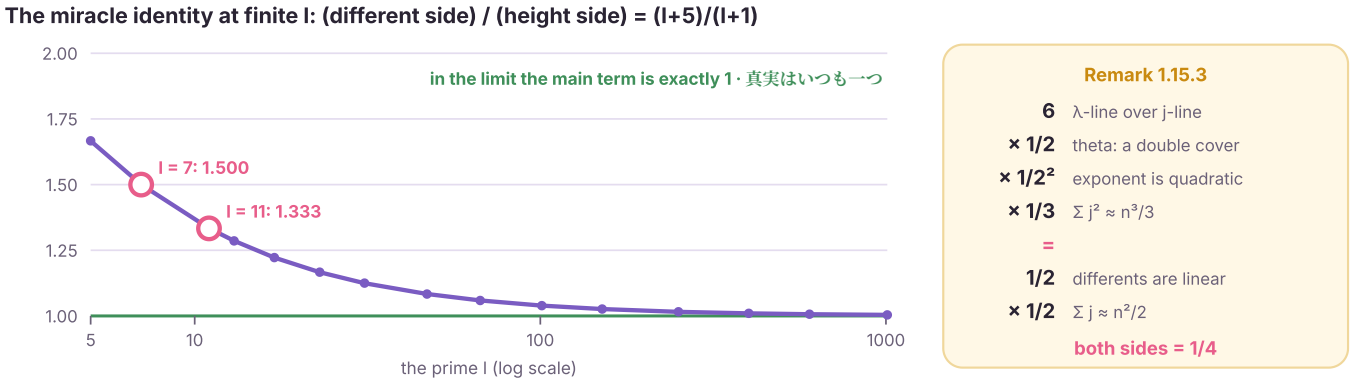


Figure 4. The ratio of the two coefficients is $(l+5)/(l+1)$: not yet 1 at the small primes our examples use (circled), and 1 in the limit. This is why the proof takes l large, just above $\sqrt{h}$, and why "the smallest l that passes Def. 3.1" is legal but beside the point.

example	l	height side $\frac{l+1}{4}$	different side $\frac{l+5}{4}$	ratio	Θ-pilot $-\frac{l+1}{24}\log q$ (Lemma 1.10)	toolkit's Θ-pilot
1 + 8 = 9	7	2.00	3.00	1.500	-1.464816	-1.464816
1 + 2400 = 2401	11	3.00	4.00	1.333	-12.101129	-12.101129
$2 + 3^{10}\cdot 109 = 23^5$	7	2.00	3.00	1.500	-20.903295	-20.903295
curve 11a1	7	2.00	3.00	1.500	-3.996492	-3.996492

The last two columns are a check on us: the survey's closed form and the number the toolkit computes by summing over j agree to six decimals for all four examples (the build fails if they do not).

Two further remarks in [Y] are worth recording. First, the identity was already visible in Mochizuki's Hodge–Arakelov theory of the late 1990s (Appendix A.3), years before there was any framework in which it could be used: "Mochizuki firstly observed this equality, and next he established the framework … in which these calculations work." Second, the 1 is sharp. Masser showed by analytic number theory that the

main term cannot be improved [Mass2], so whatever a proof of abc looks like, it has to land on this coefficient exactly.

> **OTAKU CORNER · ひとつなぎの**
> A certain pirate manga has spent more than a thousand chapters not telling us what the treasure is. IUT is kinder. The treasure is $6 \cdot \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{3} = \frac{1}{2} \cdot \frac{1}{2}$, it has been sitting in the Hodge–Arakelov papers since around 2000, and the 646 pages of [IUTchI–IV] are the voyage to an island where you are allowed to spend it.

6 Why the lattice needs infinite columns 縦は無限

Our articles say the log-theta-lattice "does not commute" and cite [IUTchIII] Remark 1.4.1. The opening of [Y] §13 gives the reason in six characters: $\log(a^N) \neq (\log a)^N$. The Θ-link is, on value groups, a power map. The log-link is a logarithm. They do not commute for the same reason they do not commute on a pocket calculator.

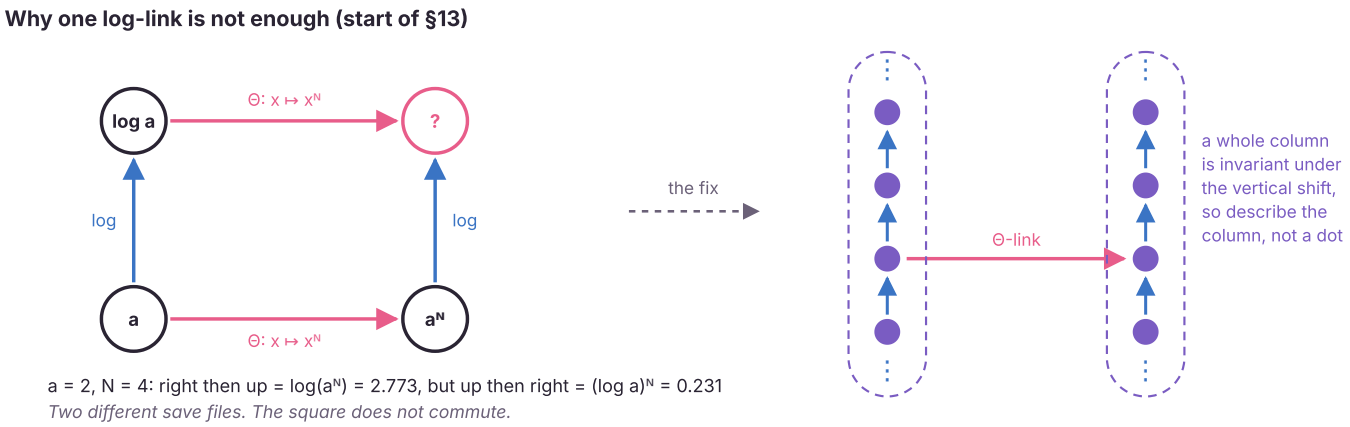


Figure 5. Left: one Θ-link and one log-link on each side form a square that does not commute, so the left log-link cannot be described from the right. Right: an entire column of log-links is unchanged when shifted up by one, so the *column* on the left can be described in terms of the column on the right.

This is the cleanest statement we have seen of why the theory needs the whole vertical line and not just one step up, and it explains the shape of the three indeterminacies. One wants to see the additive structure on the far side of a Θ-link, which only carries multiplicative data. A log-link converts one to the other, but breaks the square. Using the full column repairs the square at the cost of no longer knowing *which* row you are on, and that ignorance is precisely the vertical indeterminacy, (Indet ↑), our (Ind3).

The survey also says, concretely, how big the indeterminacies are allowed to be in the final estimate. In Lemma 1.9 the region whose volume is bounded is the holomorphic hull of the union of $\phi\left(\underline{q}^{j^2}\,\mathcal{I}\right)$ over *all* $\mathbb{Q}_p$-linear automorphisms ϕ of $\mathbb{Q}_p \otimes \mathcal{I}$ that preserve the tensor-packet log-shell $\mathcal{I}$. Our guide described (Ind2) as a compact group of isometries and (Ind1) as label permutations. Both are true, and both are swallowed in the computation by this far cruder group: anything that maps the lattice to itself. The estimate does not need to know what the indeterminacies *are*, only that they stay inside the log-shell. Remark 1.6.1 puts the underlying point memorably: once $\frac{1}{p}\mathfrak{m}_k$ is regarded as a bare $\mathbb{Z}_p$-module, nothing intrinsic distinguishes $\frac{1}{p}\varpi, \frac{1}{p}\varpi^2, \ldots, \frac{1}{p}\varpi^e$, "a kind of differentiation over $\mathbb{F}_1$".

OTAKU CORNER · 壁

Yamashita calls the Θ-link and the log-link "walls", and says Frobenius-like objects are what you build walls from while étale-like objects are what can "penetrate these walls" (§0.1, §4.3). A long-running series about a city behind three concentric walls taught a generation what happens next: the interesting characters are the ones who can get through, and the plot is about what, exactly, is on the other side. Here the answer is "another copy of the same city, with the addition table missing".

7 The banned cheat code Θ^N 禁止コマンド

The Θ-link identifies $\{\underline{q}^{j^2}\}_{1\le j\le l^*}$ on one side with $\underline{q}$ on the other. An obvious question, which Remark 13.13.3 asks and none of our write-ups did: why those exponents? If replacing $\underline{q}$ by $\underline{q}^{j^2}$ wins a factor of about l^2, replace the theta function by its N-th power and win N times more.

variant of the Θ-link	works?	why (Remark 13.13.3)
$\{\underline{q}^{j^2}\} \mapsto \underline{q}^{\lambda}, \lambda > 0$	yes, pointlessly	The left side of the final inequality becomes $\lambda \cdot 0$. For $\lambda \ll l$ nothing changes; for $\lambda > l$ the result is weaker.
$\{(\underline{q}^{j^2})^N\} \mapsto \underline{q}, N > 1$	no	All three rigidities of the mono-theta environment fail. (a) Cyclotomic rigidity uses that the commutator of the theta group is *exactly* quadratic. (b) Constant-multiple rigidity fails: the N-th power map relates non-isomorphic line bundles. (c) Θ^N has a zero of order N at the cusps, so each loop in the log-Kummer correspondence multiplies by N and it "totally collapses".
$\underline{q} \mapsto \underline{q}^{\lambda}$ **with no theta function**	no	The splitting that makes Kummer theory compatible sends $\Theta \mapsto 1$ at the zero-labelled point. The Kummer class of $\underline{q}$ is never trivial, so there is nothing to split along.

The second row has a second, independent proof that it must fail, and it is the best paragraph in the remark. If Θ^N worked, the same computation would give

$$\mathrm{ht} \;\lesssim\; \frac{1+\epsilon}{N}\,\big(\text{log-diff} + \text{log-cond}\big), \qquad N > 1,$$

which contradicts Masser's lower bound. The theory is therefore under an external constraint: its indeterminacies must be exactly as large as they are. A version of IUT that proved more would be wrong. In this sense the three rigidities of §7 of the survey are not technical conveniences; they are what stops the machine from proving a false theorem.

OTAKU CORNER · なんでやねん

Manzai needs two people: the *boke* who says the absurd thing and the *tsukkomi* who hits him with a paper fan. Remark 13.13.3 is a complete routine. Boke: "If squaring the exponent is good, why not raise the whole theta function to the power N?" Tsukkomi, with fan: "Because then abc would be false, Masser proved that in 1990, and also your Heisenberg group stopped being Heisenberg." なんでやねん。

8 Saying goodbye to scheme theory さよならスキーム論

Appendix A is five pages and is where we would now send anyone who asks what IUT is *for*. It tells the origin story in three comparison theorems: de Rham (periods like $2\pi i$), p-adic Hodge, and Mochizuki's Hodge–Arakelov theory, which evaluates sections of a line bundle at the l-torsion points and finds "Gaussian poles" $q^{j^2/8l}$ on the étale side. Then A.4 makes an argument by wishful thinking.

Suppose $E[l]$ had a global multiplicative subspace, a rank-one submodule that is the canonical μ_l at every bad place at once. Then Hodge–Arakelov theory would give an inclusion of arithmetic line bundles $\underline{q}\,\mathcal{O}_K \hookrightarrow \underline{q}^{j^2}\mathcal{O}_K$, and comparing degrees,

$$0 \;\lesssim\; -(\text{large number}) \;\approx\; -\,\mathrm{ht}\,.$$

So heights would be bounded, so there would be only finitely many elliptic curves, which is false. Hence no such subspace exists in general. A normal mathematician stops here. Yamashita's description of what Mochizuki did instead is the sentence of the paper:

> *"Instead, we respect the inclusion* $\underline{q}\,\mathcal{O}_K \hookrightarrow \underline{q}^{j^2}\mathcal{O}_K$*, and we say a good-bye to the scheme theory."*

The Θ-link *is* that inclusion, declared as an identification between two different scheme theories. It is not a contradiction because degrees in different scheme theories cannot be compared. The multiradial algorithm is what lets them be compared after all, at a price, and "we can calculate that the indeterminacies are (roughly) log-diff + log-cond". Hence $0 \lesssim -\,\mathrm{ht} + \text{log-diff} + \text{log-cond}$. In the same breath, the definition of initial Θ-data is reinterpreted: $\underline{C}_K$ and the cusp $\underline{\epsilon}$ "behave as though they are a global multiplicative subspace and a canonical generator up to $\{\pm 1\}$" (after Def. 10.1). Our toolkit stores $\underline{\epsilon}$ as the integer 1. It is in fact the ghost of an object that cannot exist.

Two more ideas from the survey's closing remarks, neither of which we had:

- **Hyperbolicity of number fields** (Remark 13.13.2). In p-adic Teichmüller theory the derivative of the canonical Frobenius lifting gives $(1-p)(2g-2) \le 0$, which says the curve is hyperbolic. The two ways of computing the q-pilot's log-volume are, by analogy, a self-intersection "$\Delta.\Delta$" of the diagonal in $\mathbb{Z} \otimes_{\mathbb{F}_1} \mathbb{Z}$, and $|\log(\Theta)| \le |\log(q)| \approx 0$ "means the hyperbolicity of number fields". The dictionary (end of §3.5): number field ↔ hyperbolic curve in characteristic p; log-link ↔ Frobenius; Θ-link ↔ the lifting $p^n/p^{n+1} \rightsquigarrow p^{n+1}/p^{n+2}$; the whole lattice ↔ a canonical lift of Frobenius.
- **Descent to** $\mathbb{F}_1$ (Remark 13.13.1). The Θ-link behaves like an "absolute Frobenius over $\mathbb{Z}$", and the three indeterminacies can be read as "descent data from $\mathbb{Z}$ to $\mathbb{F}_1$". The survey calls the multiradial algorithm "a partial (meaningful) realisation of the philosophy of 'the field of one element' $\mathbb{F}_1$".

OTAKU CORNER · 卒業

Japanese school years end in March with a graduation ceremony, a song everyone cries through, and the second button of somebody's uniform. "We say a good-bye to the scheme theory" is a graduation speech. Scheme theory was a good school. It taught us étale cohomology and how to behave around a base change. But the inclusion we want to respect lives off campus, and so, bowing once to Grothendieck-sensei, we walk out through the gate into a universe with no ring structure and a part-time job computing log-volumes.

9 What the survey does not claim 書いていないこと

A strategy guide is also useful for what it leaves out. Four things:

Uniform abc is not proved (Remark 1.15.5). The bound depends on the degree d of the field. The survey extracts the dependence the proof gives, $\mathrm{ht} \leq H_{\mathrm{unif}} \cdot \min\{1,\epsilon\}^{-3}\epsilon_d^{-3}\, d^{4+\epsilon_d} + H_{\mathcal{K}}$, and says plainly that uniformity in d "is not proved yet".

The ϵ-term is only bounded, and hints at something (Remark 1.15.4). Along the way the proof gives $\mathrm{ht} \leq \delta + *\,\delta^{1/2}\log\delta$ with $\delta = \text{log-diff} + \text{log-cond}$. Yamashita: "It seems that the exponent $\frac{1}{2}$ suggests a possible relation to Riemann hypothesis." This is offered as a suggestion, not a result.

No explicit constants. η_{prm}, ξ_{prm}, $C_{\mathcal{K}}$ and the size of Exc stay abstract. The level gauge of §4 uses the toolkit's value for η_{prm}, which comes from elsewhere.

No engagement with the dispute. The author ran seminars on IUT in 2013 and this version is dated June 2024; it proves Corollary 13.13 in under a page, by saying that after admitting the three indeterminacies the two columns' étale-like objects are isomorphic and that "the holomorphic hull of the union of possible images of Θ-pilot objects … contains a region which is isomorphic (not equal) to the region determined by the q-pilot objects". The words "isomorphic (not equal)" are exactly where the disagreement described in `multiradial_guide.pdf` §12 lives: Scholze and Stix argue the step has no content once copies are identified; Mochizuki argues that identifying them is the error; Joshi argues a needed property is unproved; the LANA formalisation reported in July 2026 that it had not yet reconstructed the required compatibility. [Y] neither mentions nor answers any of this. It is a faithful and much more readable account of what the proof says, and for that question it leaves the reader where [IUTchIII] does. We are still a toolkit, not a referee. 審判ではありません。

10 Suggested fixes, and a phrasebook 直すところ・単語帳

Items 1 to 4 were done on November 10, 2026, after this article first appeared. Item 5 followed the same day on the command line, together with two things this list did not ask for (7 and 8). Item 6 is still open, and so is everything in `iut gaps`, the toolkit's own list of what it does not compute.

#	fix	where
1 ✓	Reword the sentence in `funfun_iut.pdf` §1 to "at the bad places v_3, v_5, v_7 of $\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}$", and mention v_2 only with $\log q^{\forall}$.	`docs/funfun_iut/funfun_iut.html`
2 ✓	Carry a ramification index at bad places of at least $2l/\gcd(2l, \mathrm{ord}\, q_v)$, or print e_K as "$\geq$" instead of "1"; replace the $\log \mathfrak{d}^K$ estimate by, or print it beside, the Lemma 1.11 (1) bound.	`scripts/infographic/initialtheta-data.js`, `src/iut_core.c`
3 ✓	Change the verdict line of `ComputeThmOneTen` / `ComputeCorTwoTwo` from "holds" to something like "holds trivially; this curve is in the exceptional set ($\epsilon_E > 1$)".	`src/iut_ops_iv.c`
4 ✓	Make pipeline stage 1 item (a) use the same degree check as the validator.	`src/iut_i.c`

5 √	Add an "auto l (proof window)" choice next to "auto l": the smallest admissible prime above $\sqrt{\log q^{\forall}}$. Done as `--l auto` (conditions (P1)–(P3) of the proof of [IUTchIV] Cor 2.2); the GUI does not offer it yet.	`src/iut_ops_iv.c`: `iut_choose_l`
6	Cite [Y] in the manual and both guides; its numbering is far easier to point at than step (vii) of the proof of [IUTchIV] Theorem 1.10.	docs
7 √	Compute the bound of [Y] Prop 1.12 (= [IUTchIV] Thm 1.10) the way its proof does, one rational prime at a time with every term, and stop printing Corollary 13.13 (= Cor 3.12) with a √: it is an input, and the toolkit can only say whether its numbers are consistent with it.	`src/iut_ops_iv.c`: `iut_hull_bound_compute`
8 √	Write K_v down where that is possible. At a Tate-curve place with $p \nmid 30l$, $K_v = \mathbb{Q}_p(\zeta_{60l}, q_v^{1/30l})$ up to an unramified quadratic twist, so $e = 30l/\gcd(30l, \operatorname{ord} q_v)$ exactly and the log-shell there is no longer a model. The wild places $p \mid 30l$ still carry the "$\geq$".	`src/iut_core.c`: `iut_local_field_K`

10.1 Phrasebook: Yamashita ↔ Mochizuki ↔ this repository

[Y]	[IUTchI–IV]	toolkit / our guides	meaning
(Indet ↶↷), permutation	(Ind1)	`ind1`, "shuffled seats"	étale transport: ${}^{\dagger}G_v \cong {}^{\ddagger}G_v$, automorphisms of processions
(Indet →), horizontal	(Ind2)	`ind2Summands`, "turning the dial"	Kummer detachment along the Θ-link: ${}^{\dagger}\mathcal{O}^{\times\mu} \cong {}^{\ddagger}\mathcal{O}^{\times\mu}$
(Indet ↑), vertical	(Ind3)	`ind3`, "only fits inside"	Kummer detachment along the log-links: upper semi-compatibility
Hodge theatre	Hodge theater	`HodgeTheater`	the same; [Y] spells it the British way
l^{*}	l^{*}	`LStar`, l^{*}	$(l-1)/2$
Cor. 13.13 / Thm. 13.12	[IUTchIII] Cor. 3.12 / Thm. 3.11	stage 5	the inequality / the multiradial algorithm
Prop. 1.12 / Prop. 1.15	[IUTchIV] Thm. 1.10 / Cor. 2.2	stages 6–7	the log-volume computation / the choice of l
Def. 10.1 (1)–(7)	[IUTchI] Def. 3.1 (a)–(f)	stages 0–1	initial Θ-data; [Y] splits the conditions into seven items, not six
⊞ / ☒	⊞ / ☒	"additive" / "multiplicative"	$\mathbb{F}_l^{\times\pm}$-symmetry (units, log-shells) / $\mathbb{F}_l^{*}$-symmetry (number field)

OTAKU CORNER · コマンド入力

Mochizuki numbers his indeterminacies like a bureaucrat: (Ind1), (Ind2), (Ind3). Yamashita draws them as arrows: ↶↷, →, ↑. Anyone who grew up in a 1990s arcade recognises the second notation immediately. It is a special-move input. Rotate the stick, push forward, push up, and the Θ-pilot's log-volume comes out the other side of the screen as an upper bound. The move list in the back of the cabinet is Appendix C.6.

References 参考文献

- **[Y]** Go Yamashita, *A proof of the abc conjecture after Mochizuki*, manuscript for RIMS Kôkyûroku Bessatsu, version of June 25, 2024, 409 pages. In this repository as `pdf/A Proof of the abc Conjecture after Mochizuki (Go Yamashita, 2024-06-25).pdf`. All theorem, lemma and remark numbers without another tag refer to it.
- **[IUTchI–IV]** S. Mochizuki, *Inter-universal Teichmüller Theory I–IV*, Publ. RIMS 57 (2021). In `pdf/`.
- **[GenEll]** S. Mochizuki, *Arithmetic Elliptic Curves in General Position*, Math. J. Okayama Univ. 52 (2010). In `pdf/`.
- **[pGC]** S. Mochizuki, *The Local Pro-p Anabelian Geometry of Curves*, Invent. Math. 138 (1999).
- **[Mass2]** D. W. Masser, *Note on a conjecture of Szpiro*, Astérisque 183 (1990), 19–23.
- **[Alien], [EssLgc], [ExpEst]** as cited in `funfun_iut.pdf` and `multiradial_guide.pdf`.
- **Ours** `funfun_iut.pdf`, `multiradial_guide.pdf`, `anabelian_guide.pdf`, `docs/PIPELINE.md`, `docs/GUI.md`; example data in `docs/initial-theta-data/pipeline/`.

Built by `node scripts/gokun/build.js` from `docs/gokun_abc/gokun_abc.html`. おつかれさまでした。